ECEN 5692: HW2 Matt Hartnett

Problem 1.2.A.7:

If $b_1 = 5MHz$ then $b_2 \geq 5MHz$. Any bandwidth less than the transmit signal would lead to signal loss on the receiver.

Problem 1.2.A.8:

$x(t) = 5\sin(2\pi(f_c+7)t + 1.2)$

$I/Q \downarrow : \sqrt{2}\, x_c(t) e^{-j2\pi f_c t} * h_L(t)$

$x_c(t) = 5/2j \left(e^{j(2\pi(f_c+7)t+1.2)} - e^{-j(2\pi(f_c+7)t+1.2)} \right)$

$X(t) = \sqrt{2} \left(\frac{5}{2j} \left(e^{j(2\pi(f_c+7)t+1.2)} - e^{-j(2\pi(f_c+7)t+1.2)} \right) \left(e^{-j2\pi f_c t} \right) \right) * h_L(t)$

$= \frac{5\sqrt{2}}{2j} \left(e^{j2\pi f_c t + j2\pi 7t + 1.2j} \cdot e^{-j2\pi f_c t} - e^{-j2\pi f_c t - j2\pi 7t - 1.2j} \cdot e^{-j2\pi f_c t} \right) * h_L(t)$

$= \frac{5\sqrt{2}}{2j} \left(e^{j2\pi 7t + 1.2j} - e^{-j2\pi 2f_c t - j2\pi 7t - 1.2j} \right) * h_L(t)$

LPF distributes to both terms and since the second term is in the LPF's stopband, it is removed.

$\boxed{X(t) = 5\sqrt{2}/2j \left(e^{j2\pi 7t + 1.2j} \right)}$

Problem 1.2.A.9:

I would not expect the I/Q down converter to be $x(t) = 1$. That does not account for noise that the signal would have gained. It also assumes that the two I/Q converters are perfectly frequency and phase matched, which is unlikely.

I would expect to see $x(t) = 1 +$ random noise $+$ frequency/phase difference sinusoid.

Problem 1.2.A.10:

$x(t) = 1$

For an ideal I/Q upconverter:

$x_c(t) = R_e[x(t)\sqrt{2}\, e^{j2\pi f_c t}]$

$x_c(t) = R_e[\sqrt{2}\, e^{j2\pi f_c t}]$
$= \sqrt{2}\, e^{j2\pi f_c t}$

$y(t) = x_c(t)\sqrt{2}\, e^{-j(2\pi f_c t + \theta')} * h_L(t)$

$= 2\left(e^{j2\pi f_c t} e^{-j(2\pi f_c t + \theta')}\right) * h_L(t)$

Problem 1.2.A.10 (cont.):

$$y(t) = 2\left(e^{j2\pi f_c t} e^{-j(2\pi f_c t + \theta')}\right) * h_L(t)$$

$$= 2e^{-j\theta'} * h_L(t)$$

$$\boxed{y(t) = 2e^{-j\theta'}}$$

Problem 1.2.A.11:

$$x(t) = 1$$

$$x_c(t) = Re\left[x(t)\sqrt{2}\, e^{j2\pi f_c t}\right]$$
$$= \sqrt{2}\, e^{j2\pi f_c t}$$

$$y(t) = \left(x_c(t)\sqrt{2}\, e^{-j2\pi f_c' t}\right) * h_L(t)$$

$$= 2\left(e^{j2\pi f_c t} e^{-j2\pi f_c' t}\right) * h_L(t)$$

$$= 2\, e^{j2\pi (f_c - f_c')t} * h_L(t)$$

If we assume that the frequency difference is less than the LPF's cut off frequency:

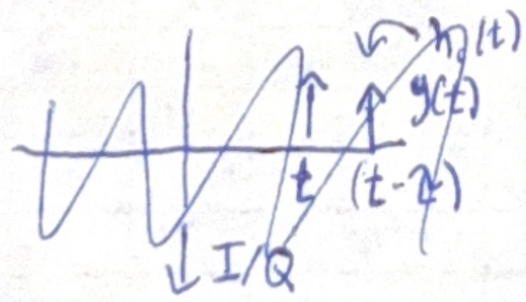
$$\boxed{y(t) = 2e^{j2\pi (f_c - f_c')t}}$$

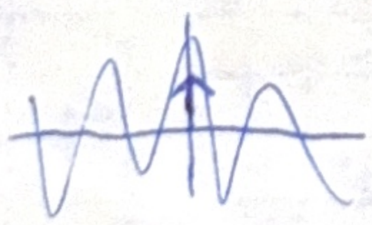
If $(f_c - f_c') > \omega_{toff}$, $y(t) = \emptyset$

Problem 1.2.A.12:

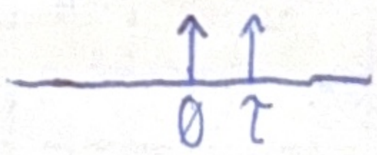
$$h_c(t) = \delta(t) + a\delta(t - \tau_1)$$

1. $h(t) = \frac{1}{\sqrt{2}}\, I/Q\downarrow(h_c(t))$





Graph of $h_c(t)$:



$$\delta(t-\tau)g(t) = \delta(t)g(\tau)$$

$$I/Q\downarrow: \sqrt{2}\, h_c(t) e^{-j2\pi f_c t} * h_L(t)$$

$$h(t) = \left(\delta(t) + a\delta(t-\tau)\right)e^{-j2\pi f_c t} * h_L(t)$$

$$\cancel{= (1 + ae^{-j2\pi f_c \tau}) * h_L(t)}$$

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Problem 1.2.A.12:

$\cancel{h(t) = (1 + ae^{-j2\pi f_c \tau}) * h_L(t)}$] ← Incorrect!

$x(t) * \delta(t - t_0) = x(t - t_0)$

$\cancel{h(t) = \delta(t)e^{-j2\pi f_c t} * h_L(t) + a\delta(t-\tau)e^{-j2\pi f_c t} * h_L(t)}$

$h(t) = \delta(t)e^{-j2\pi f_c 0} * h_L(t) + a\delta(t-\tau)e^{-j2\pi f_c \tau} * h_L(t)$

$= \delta(t) * h_L(t) + a\delta(t-\tau)e^{-j2\pi f_c \tau} * h_L(t)$

$= h_L(t) + ae^{-j2\pi f_c \tau}(h_L(t-\tau))$

$h_L(t) = \frac{\sin(\pi t b)}{\pi t}$ (sinc function)

$h(t) = \frac{\sin(\pi t 3\times10^6)}{\pi t} + 0.5e^{j\pi(-200)} \frac{\sin(\pi(t-10^{-7})3\times10^6)}{\pi(t-10^{-7})}$

$\boxed{B}$

2. $\tilde{h}(f) = \frac{1}{2}\, I/Q\downarrow(\tilde{h}_c(f))$

$\tilde{h}_c(f) = 1 + ae^{-j2\pi f \tau}$

$\tilde{h}(f) = \tilde{h}_c(f + f_c)\tilde{h}_L(f)$

$= (1 + ae^{-j2\pi(f+f_c)\tau})\, rect_b(f)$

$= \cancel{(1 + 0.5e^{}}$

$= (1 + 0.5e^{j\pi(-2f10^{-7} - 200)})\, rect_b(f)$

$\boxed{C}$